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NPS/002/018 – Pilot, Control and Telephone Cables

1. Purpose

The purpose of this document is to detail the requirements of CE Electric UK in relation to Multicore and Multipair Cables for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/018	1	Telephone Pilot and Telephone Cable

2. Scope

This Specification details the requirements for PVC insulated and sheathed multicore, armoured cables suitable for operation on cable systems having a working voltage upto and including 600/1000volts, and polyethylene insulated, polyethylene inner-sheathed "filled" and "unfilled" multipair armoured cables with PVC oversheath.

3. Technical Requirements

3.1 Multipair Cables

3.1.1 General

Polyethylene insulated, Polyethylene inner-sheathed 'filled' or 'unfilled' armoured cables with PVC oversheath, which are the intended to provide circuits for feeder protection, speech and data transmission.

3.1.2 Requirements

- Multipair cables for situations where the induced voltages will not exceed 5kV, and shall comply with the requirements of section 5 and table E2 (a) of ENATS 09-6.
- Multipair cables for situations where the induced voltages may exceed 5kV but not exceeding 15kV, and shall comply with the requirements of section 5 and table E2(b) of ENATS 09-6.

3.1.3 Conductors

Plain annealed copper wire complying with BS EN 60228:2005.

3.1.4 Conductor Insulation

Polyethylene Insulation shall be in accordance with BS 7870-8.2.

3.1.5 Conductor Size and Cable configuration

The size and numbers of cores required are as detailed in appendix 1.

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3.1.6 Cable Filling

Filled cables shall be filled with a suitable medium which shall have no detrimental effect with the core insulation and insulation colours.

3.1.7 Inner Bedding

The inner bedding shall consist of an extruded layer of a polyethylene polymer, as detailed in BS 7870-8.2.

3.1.8 Wire Armouring

The armour of the cable shall consist of one layer of galvanised steel wire as detailed in BS 7870- 8.2.

3.1.9 Identification of cores

The cores shall be clearly identified by colours in accordance with the sequence detailed in ENATS 09-6.

3.1.10 Oversheath/Embossing-Markings

The cable oversheath shall consist of an extruded covering of Black PVC type TM1 in accordance with BS EN 50363-4.1:2005. Along with the manufacturers identification mark, the sheath of the cables shall be embossed **ELECTRIC CABLE – TELE** for unfilled cables and **ELECTRIC CABLES – TELE- F** for filled cables, as per BS 7870-8.2.

3.2 PVC Insulated Multicore Cables

3.2.1 General

PVC insulated and sheathed multicore, armoured cables suitable for operation on cable systems having a working voltage upto and including 600/1000volts. The primary use is for remote operation of electrical equipment and switchgear protection circuits within substations. The cable shall comply with the requirements of section 4, table E1 of ENATS 09-6.

3.2.2 Conductor Size and Cable configuration

The size and numbers of cores required are as detailed in appendix 1.

3.2.3 Conductor

Conductors shall comply with BS EN 60228:2005 having plain annealed copper wires.

3.2.4 Core Insulation

The white PVC core insulation shall be Type TI 1 compound in accordance with BS EN 50363-3: 2005.

3.2.6 Core Identification

Cores identification shall be effected by black printed numbering on white cores the scheme of identification as detailed in ENATS 09-06.

3.2.7 Inner Bedding

The inner bedding shall consist of an extruded layer of a polymer compound having a tensile strength and elongation break of not less than 50%, thickness as detailed in BS 7870-8.1.

3.2.8 Armour Wires

The armour of the cable shall consist of one layer of galvanised steel wire as detailed in BS 7870- 8.1.

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3.2.9 Oversheath/Embossing-Markings

The cable oversheath shall consist of an extruded covering of Black PVC type TM1 in accordance with BS EN 50363-4.1:2005. Along with the manufacturer identification, the cable shall be embossed with the following legend: **ELECTRIC CABLE 600 / 1000 V AUX** as detailed in BS 7870-8.1.

4. References

4.1 External Documentation

The Multicore and Multipair cables shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

Reference	Title
ENATS 09-6 (2005)	Auxiliary Multicore and Multipair Cables
BS EN 60228:2005	Conductors of Insulated Cables
BS EN 50363:2005	Insulating, Sheathing and Covering Materials for Low Voltage Energy Cables
BS-7870: 2005	LV and MV polymeric insulated cables for use by distribution and generation utilities. Specification for multicore and multipair cables for installation above and below ground. Single wire armoured and PVC sheathed multipair cable with copper conductors. Section 8.2 Single wire armoured and PVC sheathed multipair cable with copper conductors

4.2 Internal documentation

Reference	Title
IMP/001/913	CoP for Economic Development of EHV System

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and Authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	23/09/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and Authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	29/09/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and Authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	08/10/08

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	22/10/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	17/10/08

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	20/10/08

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Appendix 1 – Requirements & Catalogue Numbers

Multipair Auxiliary Cables (Ref: CoP IMP/001/913)

Description	NEDL/YEDL Catalogue Number
7-Pair: FILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 15kV withstand induced voltage)	120861
7-Pair: UNFILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 5kV withstand induced voltage)	011585
19-Pair: FILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 5kV withstand induced voltage)	120876
19-Pair: UNFILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 5kV withstand induced voltage)	121525
37-Pair: FILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 5kV withstand induced voltage)	120880
37-Pair: UNFILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 5kV withstand induced voltage)	121597
61-Pair: FILLED : 1 x 08mm ² Copper Conductor, Polythene Insulated, Steel Wire Armoured and PVC Oversheath (up to 5kV withstand induced voltage)	120895

Multicore Auxiliary Cables (Ref: CoP IMP/001/913)

Description	NEDL/YEDL Catalogue Number
4 core: 7/0.67mm ² Copper Conductor, PVC Insulated, Steel Wire Armour and PVC Oversheath.	010801
7 Core: 7/0.67mm ² Copper Conductor, PVC Insulated, Steel Wire Armour and PVC Oversheath.	011015
12 core: 7/0.67mm ² Copper Conductor, PVC Insulated, Steel Wire Armour and PVC Oversheath.	011106
19 core: 7/0.67mm ² Copper Conductor, PVC Insulated, Steel Wire Armour and PVC Oversheath.	125507

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Appendix 2 – Addendum to Supplier Requirements

The oversheath of all cables shall be marked in accordance with the requirements of the relevant Standard specified in this document.

All cables shall be metre marked throughout the length of the cable and the start and end values shall be marked on the drum label.

Cables shall be marked with some form of reference or batch number that can be used to ensure batch trace ability of materials and manufacturing facilities used in the construction of the cable.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

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Appendix 3 – Logistical Requirements

All cable drums shall be returnable and the tenderer shall arrange to collect empty drums from the company's normal delivery locations.

Tenderers shall state at the time of tender their proposed cable drum sizes and weights for each cable type offered.

All cable drums shall be marked in accordance with the relevant cable Specification or Standard. The drum label shall also contain: -

- (a) Customers catalogue number
- (b) Name of manufacturer
- (c) Supplied length
- (d) Cable type
- (e) Number of pairs
- (f) Size of conductor
- (g) Gross and nett weights
- (h) Direction of rolling drum
- (i) The metre- marking start and end values
- (j) The unique reference number

Cable drums may be stored for long period's outdoors. All drum labels shall remain legible and durable under these conditions.

The ends of all cables shall be effectively sealed against the ingress of moisture by a method appropriate to the cable type. Tenderers shall detail at the time of tender their proposed sealing arrangement for each cable type offered.

The cable end projecting from the drum shall be protected from damage during transit, storage and handling on site.

The cable on the drum shall not be susceptible to damage during transit, storage and handling on site. Tenderers shall state at the time of tender their proposed method of protection for each cable

Each delivery length of cable shall be allocated a unique reference number. This number shall appear on the factory test sheet covering the cable length, shall be clearly marked on the drum on which the length is delivered and shall be referred to on all invoices and advice notes.

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

Pilot and telephone cables required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Type and Size)

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each type of cable offered, supplemented with a cable data sheet.

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		Clause / Requirements	Conformance Code	Remarks / Comments
	Conductors	BS EN 60228		
	Insulation			
	- Requirement	BS 7870 8.1 & 8.2: 5		
	- Voltage Test	BS 7870 8.1 & 8.2: 14.4		
	- Non Electrical	BS 7870 8.1 & 8.2: 16.2		
	- Application & Thickness	BS 7870 8.1 & 8.2: 15.3		
	- Core Identification	ENATS 09-6 Appendix F1/ 2		
	- Cable Construction	ENATS 09-6 Appendix E1/E2(a)/E2(b)		
	Bedding			
	- Layer & Thickness	BS 7870 8.1 15.6 BS 7870 8.2 15.7		
	- Testing	BS 7870 8.1 16.4 BS 7870 8.2 16.3		
	Compatibility of Filling	BS 7870 8.2 16.10		
	Water Penetration	BS 7870 8.2 15.11		
	Wire Armouring			
	- Diameter	BS7870 8.1 15.7 BS 7870 8.2 15.8		
	- Mass of Zinc Coating	BS7870 8.1 16.5 BS 7870 8.2 16.4		
	- Tensile Strength	BS7870 8.2 16.6		
	Oversheath			
	- Thickness	BS7870 8.1 15.8 BS 7870 8.2 15.9		
	- Marking	BS 7870 8.1 & 8.2: 14.7		
	- Voltage Tests	BS 7870 8.1 & 8.2: 14.6		
	- Compatibility	BS 7870 8.1 16.8		
	- Tests	BS 7870 8.1 & 8.2: 16.7		